

Puiseux series about exceptional singularities dictated by symmetry-allowed Hessenberg forms of perturbation matrices

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We develop a systematic framework for determining the nature of exceptional points of n th order (EP_n s) in non-Hermitian (NH) systems, represented by complex square matrices. By expressing symmetry-preserving perturbations ($\propto \epsilon$) in the Jordan-normal basis of the defective matrix at an EP_n , we show that the upper- ℓ Hessenberg structure of the perturbation matrix directly dictates the leading-order eigenvalue- and eigenvector-splitting to be $\propto \epsilon^{1/\ell}$, when expanded in a Puiseux series. Applying this to three-band NH models invariant under parity (P), charge-conjugation (C), or parity-time-reversal (PT), we find that EP_3 s in P- and C-symmetric systems are restricted to at most $\sim \epsilon^{1/2}$ branch points, while PT-symmetric systems generically support EP_3 s with the strongest possible singularities (*viz.*, $\sim \epsilon^{1/3}$). We illustrate these results with concrete three-dimensional models in which exceptional curves and surfaces emerge. We further show that fine-tuned perturbations can suppress the leading-order branch point to a less-singular splitting, which have implications for designing direction-dependent EP-based sensors. The Appendices extend the analysis to four-band P- and C-symmetric models, establishing the existence of EP_4 s with $\sim \epsilon^{1/4}$ and $\sim \epsilon^{1/2}$ singularities, respectively.

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I. INTRODUCTION

Non-Hermitian (NH) topological phases have become a central topic in condensed matter physics, following the realization that stable band-crossing points are more generic and abundant than those found in Hermitian systems [1–5]. The robust topology of such stable NH phases [3] is characterized by invariance under various symmetries of complex matrices [4,6], strongly hinged on the mathematical concept of exceptional points (EPs). EPs are singular points in the parameter space of complex matrices at which two or more eigenvalues coalesce, sharing a single linearly independent eigenvector [7–11]. EPs can be understood as the natural dissipative analogs of the stable nodal points [7,8] familiar from the Weyl nodes, which are the well-known Hermitian systems serving as the poster child of the concept of topology realized in solid-state systems. The emergence of EPs, in fact, gives rise to a rich array of topologically robust phenomena in generic NH scenarios, examples including bulk Fermi arcs [12] and unidirectional lasing [13]. All these have propelled the understanding of EPs to an interdisciplinary research frontier, with their relevance spanning classical metamaterials and quantum condensed matter systems [10]. In this context, it is worthwhile to mention that the intimate connection of EPs with topological phase transitions in diverse contexts has been firmly established in our current scientific knowledge, as they are invoked in widely different studies [3–5,14–19].

While the initial studies involving spectral singularities of *NH Hamiltonians* were directed toward second-order EPs (EP_2 s), which arise naturally in two-dimensional (2D) NH

systems [12,20], the abundance of higher-order EPs (e.g., third-order EPs) has been demonstrated in Ref. [4]. Exceptional curves and surfaces of second- and higher-order have also been identified under relevant circumstances [21–24]. From a general perspective, realizing an n th order EP (which we label as EP_n) requires the tuning of $(2n - 2)$ real parameters, such as components of the lattice-momentum vector. This counting argument implies that EPs of the order $n \geq 3$ can only be expected in at least four-dimensional systems. Nevertheless, we have established that generic NH symmetries can stabilize EP_3 s in 2D systems. In particular, this implies that three-dimensional (3D) systems will host such exceptional singularities in the form of nodal curves, which we call exceptional curves (ECs). In this paper, we continue the ongoing efforts to characterize the endless avatars of exceptional singularities by digging into how the degree of the singularity originates from the overall structure of the complex matrix. Since the EPs are associated with branch-point singularities, an expansion about an EP would be dictated by a Puiseux series [25–27], which is a generalized power series allowing for negative and fractional exponents of the indeterminate. In particular, we study three-band NH models, which are represented by 3×3 complex matrices invariant under parity (P), charge conjugation (C), or parity-time reversal (PT).

The paper is organized as follows: In Sec. II, we explain the mathematical objects and concepts needed to understand the nature of possible perturbations to an $s \times s$ complex matrix, which can be taken to correspond to an s -band NH model in the context of solid-state systems. In Sec. III, we review and apply the idea of perturbations in the form of Hessenberg matrices to a 2×2 matrix, which produces the simplest-possible singular degeneracies, namely, EP_2 s. Section IV is devoted to deriving the generic forms of the singularities for

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three-band models under the protection of three different kinds of symmetries, spanning P, C, and PT symmetries. There, we supplement our analysis of generic matrices with specific 3D systems in the momentum space. We end with some discussions and outlook in Sec. V. The Appendices deal with analyzing C- and P-symmetric four-band models.

II. PERTURBATION AROUND AN EXCEPTIONAL DEGENERACY

In this section, we provide the mathematical definitions required to understand the singularity structure of exceptional degeneracies for generic complex matrices. An $s \times s$ matrix which has $(\ell - 1)$ subdiagonals, adjacent to the principal diagonal in the lower half, which do not identically comprise zero elements, is referred to as an upper- ℓ Hessenberg matrix, $B_{s,\ell}$. In other words, if the matrix elements are denoted as $m_{i,j}$, then we have $m_{i,j} = 0$ for $i > j + \ell - 1$. In this language, the common nomenclature of an *upper Hessenberg matrix* is actually an upper-2 Hessenberg matrix. Let us consider a Jordan block, $\mathcal{J}_s(\lambda)$, of dimension s and s -fold degenerate eigenvalue λ . By examining the perturbed matrix $\mathcal{J}_s + \epsilon B_{s,\ell}$, Ref. [26] has derived how the degenerate eigenvalues of $\mathcal{J}_s$ split under a perturbation of the form $\epsilon B_{s,\ell}$. The main result can be summarized as the nature of the eigenvalues ($\{\lambda_{i,n_{s_i}}\}$) of all possible perturbed matrices and can be grouped in the following manner, characterized by the respective Puiseux series:

- (i) Prepare all possible sets of $\{s_i\}$, such that $\sum_i s_i = s$.
- (ii) For each i , we have a ring of size s_i , defined by the elements,

$$\lambda_{i,n_{s_i}} = \lambda + \alpha_{i,1} \omega_{s_i}^{n_{s_i}} \epsilon^{\frac{1}{s_i}} + \alpha_{i,2} \omega_{s_i}^{2n_{s_i}} \epsilon^{\frac{2}{s_i}} + \dots, \quad \omega_{s_i} = e^{\frac{2\pi i}{s_i}}, \quad \text{for } n_{s_i} \in [1, s_i]. \quad (1)$$

Let us illustrate this for some values of s .

- (i) $s = 2$.

We can have two sets, *viz.*, $\{s_1 = 1, s_2 = 1\}$ and $\{s_1 = 2\}$. For set 1, we have $\omega_{s_1} = \omega_{s_2} = \omega_1 \equiv 1$ and the eigenvalues behaving as

$$\begin{aligned} \lambda_{1,1} &= \lambda + \alpha_{1,1} \epsilon + \alpha_{1,2} \epsilon^2 + \dots \\ \text{and } \lambda_{2,1} &= \lambda + \alpha_{2,1} \epsilon + \alpha_{2,2} \epsilon^2 + \dots \end{aligned} \quad (2)$$

These give rise to two rings of the trivial size of unity. For set 2, we have $\omega_{s_1} = \omega_2 \equiv -1$ and the eigenvalues behaving as

$$\begin{aligned} \lambda_{1,1} &= \lambda - \alpha_{1,1} \epsilon^{\frac{1}{2}} + \alpha_{1,2} \epsilon + \dots \\ \text{and } \lambda_{1,2} &= \lambda + \alpha_{1,1} \epsilon + \alpha_{1,2} \epsilon^2 + \dots \end{aligned} \quad (3)$$

These constitute one ring of size 2.

- (ii) $s = 3$.

We can have three sets, *viz.*, $\{s_1 = 1, s_2 = 1, s_3 = 1\}$, $\{s_1 = 2, s_2 = 1\}$, and $\{s_1 = 3\}$.

For set 1, we have $\omega_{s_1} = \omega_{s_2} = \omega_{s_3} = \omega_1 \equiv 1$ and the eigenvalues behaving as

$$\begin{aligned} \lambda_{1,1} &= \lambda + \alpha_{1,1} \epsilon + \alpha_{1,2} \epsilon^2 + \dots, \\ \lambda_{2,1} &= \lambda + \alpha_{2,1} \epsilon + \alpha_{2,2} \epsilon^2 + \dots, \\ \text{and } \lambda_{3,1} &= \lambda + \alpha_{3,1} \epsilon + \alpha_{3,2} \epsilon^2 + \dots \end{aligned} \quad (4)$$

These give rise to three rings of the trivial size of unity. For set 2, we have $\omega_{s_1} = \omega_2 \equiv -1$, $\omega_{s_2} = \omega_1 \equiv 1$, and the eigenvalues behaving as

$$\begin{aligned} \lambda_{1,1} &= \lambda - \alpha_{1,1} \epsilon^{\frac{1}{2}} + \alpha_{1,2} \epsilon + \dots, \\ \lambda_{2,1} &= \lambda + \alpha_{1,1} \epsilon^{\frac{1}{2}} + \alpha_{1,2} \epsilon + \dots, \\ \text{and } \lambda_{3,1} &= \lambda + \alpha_{2,1} \epsilon + \alpha_{2,2} \epsilon^2 + \dots \end{aligned} \quad (5)$$

These constitute one ring of size 2 and one trivial ring of size unity. For set 3, we have $\omega_{s_3} = \omega_3 \equiv e^{\frac{2\pi i}{3}}$ and the eigenvalues behaving as

$$\begin{aligned} \lambda_{1,1} &= \lambda + \alpha_{1,1} \omega_3 \epsilon^{\frac{1}{3}} + \alpha_{1,2} \omega_3^2 \epsilon^{\frac{2}{3}} + \alpha_{1,3} \omega_3^3 \epsilon + \dots, \\ \lambda_{1,2} &= \lambda + \alpha_{1,1} \omega_3^2 \epsilon^{\frac{1}{3}} + \alpha_{1,2} \omega_3^4 \epsilon^{\frac{2}{3}} + \alpha_{1,3} \omega_3^6 \epsilon + \dots, \\ \text{and } \lambda_{1,3} &= \lambda + \alpha_{1,1} \omega_3^3 \epsilon^{\frac{1}{3}} + \alpha_{1,2} \omega_3^6 \epsilon^{\frac{2}{3}} + \alpha_{1,3} \omega_3^9 \epsilon + \dots \end{aligned} \quad (6)$$

These constitute one ring of size 3.

We emphasize that, for $\epsilon \neq 0$, the perturbed matrix is generically nondegenerate and, hence, is no longer an EP in the strict sense of exact degeneracy plus defectiveness—the terminology “singular” refers instead to $\epsilon = 0$ remaining a branch point of a multivalued function the multivalued function $\lambda(\epsilon)$, whose Puiseux expansion starts at the fractional power $\epsilon^{\frac{1}{k}}$ rather than at ϵ^1 .

III. REVIEW OF EXCEPTIONAL POINTS IN TWO-BAND MODELS

For two-band models, we can get only EPs of the lowest-possible order, *viz.*, EP₂, because a perturbation in the form of a Hessenberg matrix can only be of the upper-2 type. Nevertheless, we demonstrate this point explicitly by taking an example. Consider the non-Hermitian matrix

$$\mathcal{H}_2 = \begin{bmatrix} 0 & a \\ 1 & 0 \end{bmatrix}. \quad (7)$$

For $a = 0$, it reduces to a singular matrix hosting an EP₂ with the degenerate eigenvalue $\lambda = 0$. Its Jordan-normal form is

$$S^{-1} \mathcal{H}_2 S = \mathcal{J}_2(0), \quad \text{with } S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad (8)$$

Adding a perturbation $\epsilon B_{2,2}$, with

$$B_{2,2} = \begin{bmatrix} 0 & 0 \\ \epsilon & 0 \end{bmatrix}, \quad \text{where } S B_{2,2} S^{-1} = \begin{bmatrix} 0 & \epsilon \\ 0 & 0 \end{bmatrix}, \quad (9)$$

the eigenvalues of the perturbed matrix, $\mathcal{H}_2 + \epsilon S B_{2,2} S^{-1}$ (or, equivalently, $\mathcal{J}_2 + \epsilon B_{2,2}$) are given by $\pm\sqrt{\epsilon}$. Thus the generic behavior around an EP₂ is that around a square-root branch point.

IV. EXCEPTIONAL SINGULARITIES IN THREE-BAND MODELS

In this section, our aim is to explore the emergence of higher-order EPs in three-band models, because they are able to host EP₃s [4]. We show how symmetry restrictions dictate the leading-order singular nature of these EPs, focusing on P-, C-, and PT-symmetry classes [6]. Depending on the symmetry

class, the EPs might show cube-root or square-root branch points. We also provide some examples, falling into one of these classes, obtained from NH generalizations of 3D band structures.

A. P-symmetric systems

The imposition of the P symmetry to a 3×3 non-Hermitian matrix leads to the form [4]

$$\mathcal{H}_3^P = \begin{bmatrix} 0 & b & c \\ d & 0 & 0 \\ g & 0 & 0 \end{bmatrix}, \quad P = \text{diag}\{-1, 1, 1\},$$

$$P \mathcal{H}_3^P P^{-1} = -\mathcal{H}_3^P. \quad (10)$$

The 3×3 matrix P forms a three-dimensional realization of the P symmetry. We note that, an unbroken P symmetry constrains the set of eigenvalues, $\{E_i\}$, to obey the condition $\{E_i\} = \{-E_j\}$ [1]. Consequently, for an odd-dimensional square matrix, there is always one zero eigenvalue (i.e., flat or nondispersive band). For $b = -c g/d$, it hosts an EP₃ with the threefold-degenerate eigenvalue $\lambda = 0$. At that point, its Jordan-normal form is

$$S^{-1} \mathcal{H}_{\text{EP}}^P S = \mathcal{J}_3(0), \quad \text{with } S = \begin{bmatrix} 0 & \frac{1}{g} & 0 \\ \frac{d}{g} & 0 & -\frac{d}{cg^2} \\ 1 & 0 & 0 \end{bmatrix}$$

and $\mathcal{H}_{\text{EP}}^P = \mathcal{H}_3^P|_{b=-cg/d}$. (11)

Adding a perturbation $\epsilon \tilde{B}_{3,2}^P$ to $\mathcal{H}_{\text{EP}}^P$, with

$$\tilde{B}_{3,2}^P = \begin{bmatrix} 0 & a_1 & a_2 \\ a_3 & 0 & 0 \\ a_4 & 0 & 0 \end{bmatrix}, \quad \text{where}$$

$$S^{-1} \tilde{B}_{3,2}^P S = \begin{bmatrix} 0 & \frac{a_4}{g} & 0 \\ a_1 d + a_2 g & 0 & -\frac{a_1 d}{cg} \\ 0 & c(a_4 - \frac{a_3 g}{d}) & 0 \end{bmatrix}, \quad (12)$$

the three eigenvalues are obtained as $\{0, \pm \varepsilon_2\}$, where $\varepsilon_2 = \sqrt{c(a_4 - \frac{a_3 g}{d}) + a_1(a_3 \epsilon + d) + a_2(a_4 \epsilon + g)} \epsilon^{1/2}$. This conforms to the generic form shown in Eq. (5). We note that

$$(0, \pm \{a e^{-i\alpha} a_3 \sin \theta + a_1(a_3 \epsilon + a e^{i\alpha} \sin \theta) - i a a_4 (\cos \theta - 1) + a_2[a_4 \epsilon + i a (\cos \theta + 1)]\} \epsilon^{1/2}).$$

If we set $a_4 = \frac{a_2(\cos \theta + 1) - i e^{-i\alpha}(a_3 + e^{2i\alpha} a_1) \sin \theta}{\cos \theta - 1}$, the nonzero eigenvalues behave as $\sim \epsilon$, showing that the coefficient of the $\epsilon^{1/2}$ term is adjusted to vanish for this choice of parameters.

Two parent lattice versions of the continuum model of Eq. (13) are given by

$$\mathcal{H}_3^{\text{hopf1}} = \begin{bmatrix} 0 & \sin k_x - i \sin k_y & \Delta + e^{-ik_z} + \cos k_x + \cos k_y + i a \\ \sin k_x - i \sin k_y & 0 & 0 \\ \Delta + e^{ik_z} + \cos k_x + \cos k_y + i a & 0 & 0 \end{bmatrix}$$

and $\mathcal{H}_3^{\text{hopf2}} = \begin{bmatrix} 0 & w_k & -i \sin k_z + \frac{2ia}{3} \\ w_k^* & 0 & 0 \\ i \sin k_z + \frac{2ia}{3} & 0 & 0 \end{bmatrix},$

where $w_k = \frac{2}{3} \sum_{j=1}^3 e^{ik \cdot \delta_j}$, $\delta_1 = \frac{1}{2}(\sqrt{3}, 1, 0)$, $\delta_2 = \frac{1}{2}(-\sqrt{3}, 1, 0)$, $\delta_3 = (0, -1, 0)$. (14)

a P-symmetry-preserving perturbation takes the form of an upper-2 Hessenberg matrix, $B_{3,2}$, in the Jordan-normal basis of the defective matrix, $\mathcal{H}_{\text{EP}}^P$. And as commented above, the flat band remains unaffected as long the P-symmetry remains unbroken. In particular, an upper-3 Hessenberg matrix, $B_{3,3}$, would break the P-symmetric form, although it could produce a leading-order splitting $\sim \epsilon^{1/3}$. One more crucial observation is that, upon setting $a_4 = a_3 g/d - (a_1 d + a_2 g)/c$, the leading-order behavior of ε_2 reduces to $\sim \epsilon$, since the coefficient of the square-root term vanishes. The directional dependence discussed here should not be interpreted as a change in the nature of the underlying EP. The EP₃ remains intact, since the defective Hamiltonian at $\epsilon = 0$ continues to possess a single $\mathcal{J}_3(0)$ Jordan block. Tuning parameters such as a_4 selects a special symmetry-preserving perturbation direction, for which the coefficient of the leading $\epsilon^{1/2}$ term in the Puiseux expansion vanishes. The splitting is then governed by the next nonvanishing contribution, resulting in a linear dependence on ϵ . This does not correspond to a reduction to an EP₂ or to a Dirac point; instead, it represents a direction-dependent linear dispersion away from the same EP₃. The fine-tuned directions form a codimension-one subset within the symmetry-allowed perturbation space, while generic directions recover the usual $\epsilon^{1/2}$ behavior.

Let us now consider a physical example in the form of the continuum model for a node carrying a dipole of Berry curvature (BC), which constitutes a three-band Hopf semimetal [28–31]. A simple generalized NH version can be written as

$$\mathcal{H}_3^{\text{hopf}} = \begin{bmatrix} 0 & k_x - i k_y & -i k_z + i a \\ k_x - i k_y & 0 & 0 \\ i k_z + i a & 0 & 0 \end{bmatrix}, \quad (13)$$

obeying $P \mathcal{H}_3^{\text{hopf}} P^{-1} = -\mathcal{H}_3^{\text{hopf}}$. The non-Hermiticity appears for a nonzero value of a , which we assume to be real here. The system has a continuous surface of exceptional singularities of order 3 (which we label as ES₃) on the sphere, $k_x^2 + k_y^2 + k_z^2 = a^2$. The ES₃ is characterized by the Jordan-normal form $\mathcal{J}_3(0)$. Let the ES₃ be parametrized by $k_x = a \sin \theta \cos \phi$, $k_y = a \sin \theta \sin \phi$, and $k_z = a \cos \theta$. A P-symmetry-preserving perturbation of the form of $\epsilon \tilde{B}_{3,2}^P$, with respect to the singular matrix at the location of the ES₃, produces the eigenvalues,

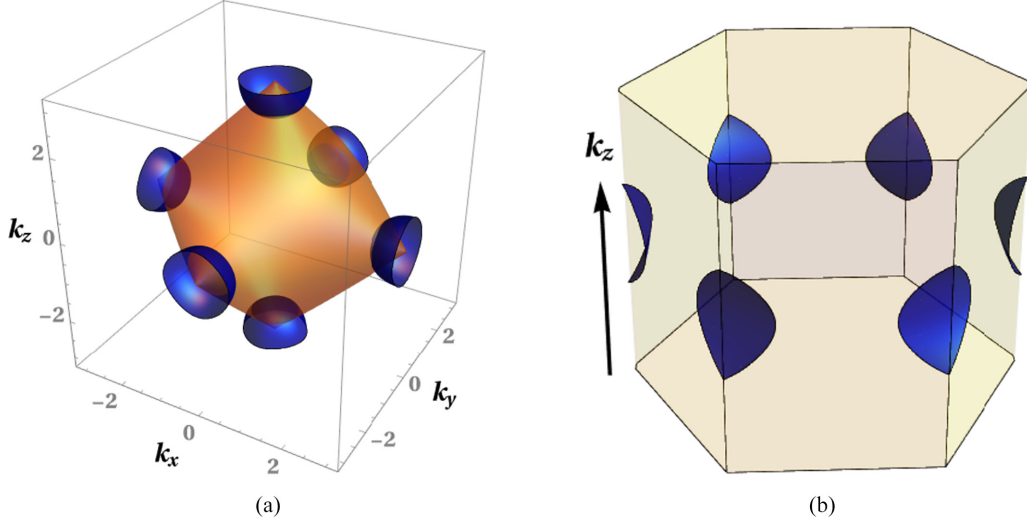


FIG. 1. Third-order exceptional degeneracies for the lattice models represented by Eq. (14). (a) Three pairs of EC_3 curves show up upon setting $\Delta = -1$ and $a = 0.75$ in $\mathcal{H}_3^{\text{hopf1}}$. (b) Three pairs of exceptional surfaces show up upon setting $a = 1$ in $\mathcal{H}_3^{\text{hopf2}}$.

The eigenvalues are $\{0, \pm\sqrt{f_1 + if_2}\}$ and $\{0, \pm\sqrt{f_3}\}$ for the two systems, respectively, where

$$\begin{aligned} f_1 &= 3 + \Delta^2 + 2 \cos k_x (\Delta + \cos k_y + \cos k_z) + 2\Delta(\cos k_y + \cos k_z) + 2 \cos k_y \cos k_z - a^2, \\ f_2 &= 2a(\Delta + \cos k_x + \cos k_y + \cos k_z), \\ f_3 &= \frac{4}{9} \left[3 + 4 \cos \left(\frac{\sqrt{3}k_x}{2} \right) \cos \left(\frac{3k_y}{2} \right) + 2 \cos(\sqrt{3}k_x) - a^2 \right] + \sin^2 k_z. \end{aligned} \quad (15)$$

For the original Hermitian version with $a = 0$, $\mathcal{H}_3^{\text{hopf1}}$ corresponds to systems harboring an odd number of nodal points harboring BC dipoles in the Brillouin zone (BZ), upon setting Δ to ± 1 or ± 3 [28]. For the case of $\Delta = -1$, these nodal points appear at the points $(\pi, 0, 0)$, $(0, \pi, 0)$, and $(0, 0, \pi)$ in the BZ. The effect of adding a nonzero a is that each nodal point transfigures into an exceptional curve of third-order (let us call it EC_3), which is the intersection of the two surfaces, $f_1 = 0$ and $f_2 = 0$. An example of this behavior is illustrated in Fig. 1(a), where we have set $\Delta = 1$ and $a = 0.75$ —we get three pairs of EC_3 curves.

For the original Hermitian version with $a = 0$, $\mathcal{H}_3^{\text{hopf2}}$ corresponds to systems harboring an even number of nodal points harboring BC dipoles in a tetragonal BZ, located at $k_z = 0$ [28]. Because the Bravais period gets doubled along the $\hat{z}$ -direction, the k_z values run from $-\pi/2$ to $\pi/2$ (i.e., its range gets halved). Thus a node can appear only for $k_z = 0$. The $k_x k_y$ -plane at each k_z forms a 2D slice in the shape of a hexagonal BZ (hBZ) and, basically, the dispersion is identical to the dispersion of graphene when Dirac points emerge at the K and K' points of the hBZ. If we take the entire 3D picture, there are inequivalent BC dipoles at K and K' , because the dipole-axes' orientations are opposite (i.e., along $\pm \hat{k}_z$). Upon adding a nonzero a , each dipole transfigures into an ES when $\sqrt{f_3}$ transitions from purely real to purely imaginary—thus, a regular nodal-point degeneracy blows up into a threefold-degenerate eigenvalue surface, displaying square-root singularity. We illustrate this via an example in Fig. 1(b) by choosing $a = 1$ —we observe three pairs of ES_3 s.

B. C-symmetric systems

Let us consider a C-symmetric non-Hermitian matrix,

$$\mathcal{H}_3^C = \begin{bmatrix} d & b & 0 \\ c & 0 & b \\ 0 & c & -d \end{bmatrix}, \quad C = \text{diag}\{1, -1, 1\},$$

$$C(\mathcal{H}_3^C)^T C^{-1} = \epsilon_c \mathcal{H}_3^C \quad \text{with } \epsilon_c = -1. \quad (16)$$

Here, $\{b, c, d\}$ are independent complex numbers. Just like the P-symmetric case, the C symmetry with $\epsilon_c = -1$ constrains the set of eigenvalues, $\{E_i\}$, to obey the condition $\{E_i\} = \{-E_j\}$ [1]. This implies that there will always be one zero eigenvalue (i.e., flat or nondispersive band) for three-band systems. For $d = \pm i\sqrt{2bc}$, it hosts an EP_3 with a threefold-degenerate eigenvalue, $\lambda = 0$. Picking the value $d = i\sqrt{2bc}$, the Jordan-normal form at this EP_3 is

$$S^{-1} \mathcal{H}_{EP}^C S = \mathcal{J}_3(0), \quad \text{with } S = \begin{bmatrix} -\frac{b}{c} & \frac{i\sqrt{2bc}}{c^2} & \frac{1}{c^2} \\ \frac{i\sqrt{2b}}{\sqrt{c}} & \frac{1}{c} & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

and $\mathcal{H}_{EP}^C = \mathcal{H}_3^C|_{d=i\sqrt{2bc}}$. (17)

We add a C-symmetry-preserving perturbation, $\epsilon \tilde{B}_{3,2}^C$, to $\mathcal{H}_{EP}^C$, with

$$\tilde{B}_{3,2}^C = \begin{bmatrix} a_1 & a_2 & 0 \\ a_3 & 0 & a_2 \\ 0 & a_3 & -a_1 \end{bmatrix}, \quad \text{where } S^{-1} \tilde{B}_{3,2}^C S = \begin{bmatrix} -a_1 + \frac{i\sqrt{2ba_3}}{\sqrt{c}} & \frac{a_3}{c} & 0 \\ i\sqrt{2bca_1} + a_3b + a_2c & 0 & \frac{a_3}{c} \\ 0 & i\sqrt{2bca_1} + a_3b + a_2c & a_1 - \frac{i\sqrt{2ba_3}}{\sqrt{c}} \end{bmatrix}. \quad (18)$$

The three eigenvalues are obtained as $\{0, \pm \epsilon_2\}$, where $\epsilon_2 = \sqrt{2i\sqrt{2bca_1} + 2a_3b + 2a_2(a_3\epsilon + c) + a_1^2\epsilon} \epsilon^{1/2}$. This conforms to the generic form shown in Eq. (5). We note that a C-symmetry-preserving perturbation takes the form of an upper-2 Hessenberg matrix, $B_{3,2}$, in the Jordan-normal basis of the defective matrix, $\mathcal{H}_{EP}^C$. In particular, an upper-3 Hessenberg matrix, $B_{3,3}$, would break the C-symmetric form, although it could produce a leading-order splitting, $\sim \epsilon^{1/3}$. One more crucial observation is that, upon setting $a_1 = i(a_3b + a_2c)/\sqrt{2bc}$, the leading-order behavior of ϵ_2 reduces to $\sim \epsilon$, since the coefficient of the square-root term vanishes.

Let us now demonstrate the consequences using a specific example in the form of the continuum model for a pseudospin-1 node [32–34]. A simple generalized NH version can be written as

$$\mathcal{H}_3^{\text{spin1}} = \begin{bmatrix} k_z + ia & \frac{k_x - ik_y}{\sqrt{2}} & 0 \\ \frac{k_x + ik_y}{\sqrt{2}} & 0 & \frac{k_x - ik_y}{\sqrt{2}} \\ 0 & \frac{k_x + ik_y}{\sqrt{2}} & -k_z - ia \end{bmatrix}, \quad (19)$$

obeying $C[\mathcal{H}_3^{\text{spin1}}]^T C^{-1} = -\mathcal{H}_3^{\text{spin1}}$ for the realization of the symmetry shown in Eq. (16). The non-Hermiticity appears for a nonzero value of a , which we take to be real here, with eigenvalues equaling $\{0, \pm \sqrt{k_x^2 + k_y^2 + k_z^2 - a^2 + 2iak_z}\}$. The system hosts an EP₃ when $k_x = a \cos \alpha$, $k_y = a \sin \alpha$, and $k_z = 0$, characterized by the threefold-degenerate eigenvalue of zero and a single linearly independent eigenvector $\propto [-1, i\sqrt{2}e^{i\alpha}, e^{2i\alpha}]^T$. The singularity is actually an EC₃ (here, a circle), as illustrated in Fig. 2, and the Jordan-normal

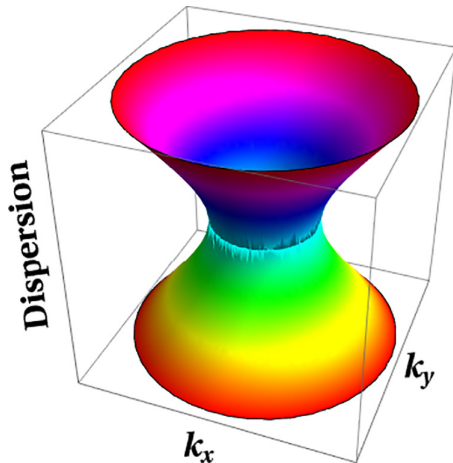


FIG. 2. A third-order exceptional circle (EC₃) for a three-band model, exemplified by Eq. (19), showing square-root singularities.

form of the defective matrix is given by $\mathcal{J}_3(0)$. A C-symmetry-preserving perturbation of the form of $\epsilon \tilde{B}_{3,2}$, with respect to the location of the EC₃, produces the eigenvalues $\{0, \pm \sqrt{\sqrt{2}ae^{-i\alpha}(a_3 + e^{2i\alpha}a_2) + (a_1^2 + 2a_2a_3)\epsilon + 2iaa_1} \epsilon^{1/2}\}$.

C. PT-symmetric systems

In Ref. [4] (see also the Supplemental Material therein), we have derived the generic form of a three-band NH matrix obeying PT symmetry. It has the form of

$$H_3^{\text{PT}} = \begin{bmatrix} \text{real}_1 & i\text{real}_2 & \text{real}_3(1-i) \\ i\text{real}_4 & \text{real}_5 & \text{real}_6(1+i) \\ \text{real}_7(1+i) & \text{real}_8(1-i) & \text{real}_9 \end{bmatrix}, \quad (20)$$

which obeys the constraint $H_3^{\text{PT}} = (PK)(H_3^{\text{PT}})^*(PK)^{-1}$, with $P = \text{diag}\{1, -1, 1\}$ and $K = \text{diag}\{1, 1, i\}$. As conveyed by the nomenclature, the nine independent parameters $\{\text{real}_1, \text{real}_2, \text{real}_3, \text{real}_4, \text{real}_5, \text{real}_6, \text{real}_7, \text{real}_8, \text{real}_9\}$ are all real. The characteristic polynomial is of cubic order, which can be used to find the eigenvalues using the Cardano's method.

A specific linear-in-momentum model, which conforms to the above PT-symmetric form and exhibits EC₂s and EP₃s, has been provided in Ref. [4]. Here, we work with a 3D version, captured by

$$\mathcal{H}_3^{\text{PT}} = \begin{bmatrix} k_z - a & ik_x & (-1+i)a \\ -ik_x & 0 & (1+i)k_y \\ (1+i)a & (1-i)k_y & a - k_z \end{bmatrix}. \quad (21)$$

The matrix is NH for $a \neq 0$. Defining the variables $\omega = \frac{-1+\sqrt{3}i}{2}$ and $\alpha_{\pm} \equiv \sqrt[3]{q \pm \sqrt{p^3 + q^2}}$, the three eigenvalues turn out to be

$$\{E_1, E_2, E_3\} = \{\alpha_+ + \alpha_-, \omega\alpha_+ + \omega^*\alpha_-, \omega^*\alpha_+ + \omega\alpha_-\},$$

$$\text{where } p = \frac{k_x^2 + 2k_y^2 + k_z^2 - a^2 - 2ak_z}{3} \quad \text{and}$$

$$q = \frac{(k_x^2 - 2k_y^2)(a - k_z)}{2}. \quad (22)$$

The real-valued function $p^3 + q^2$ plays a key role as surfaces hosting EP₂s appear at $p^3 + q^2 = 0$ (let us label the singular surface as ES₂). Additionally, the curve where the surfaces for $p = 0$ and $q = 0$ vanish forms an EC₃, characterized by the Jordan-normal form $\mathcal{J}_3(0)$. Figure 3 shows the nature of the various surfaces and the knotted EC₃ curve for $a = 1.25$.

Here, PT symmetry allows a perturbation of the form of $B_{3,3}$, thus allowing cube-root singularities. Instead of trying to write down the complicated expressions of the eigenvalues for such generic perturbations, we focus on two interesting

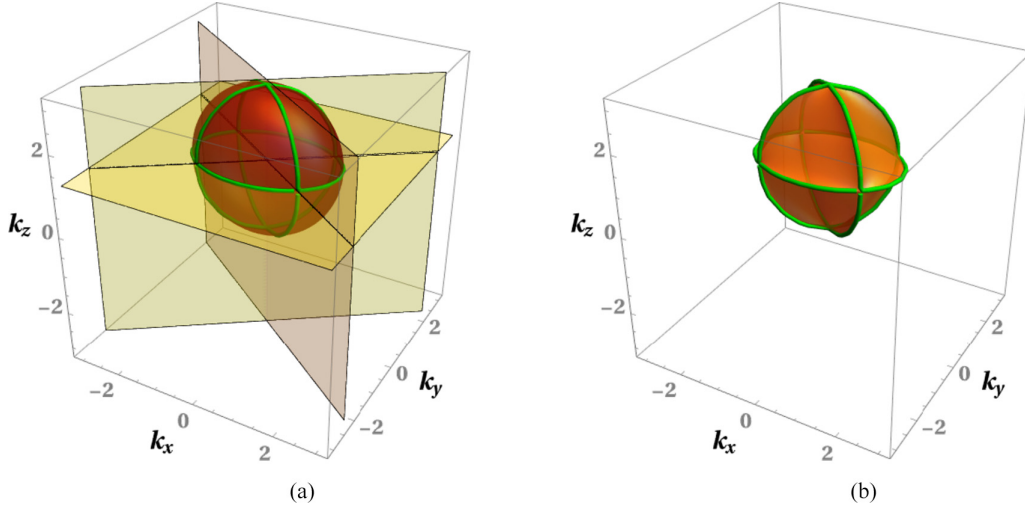


FIG. 3. Using $a = 1.25$ in Eq. (22): (a) The surfaces arising from $p = 0$ (red) and $q = 0$ (light yellow), which intersect to give a knotted curve (green) representing EC_3 . (b) The surface of ES_2 representing $p^3 + q^2 = 0$ (orange), on which the curve of EC_3 (green) is superimposed.

cases when the singularities can be identified easily via simple analytical expressions.

(i) For $k_z = 0$, we have $p = 0$ along the ellipse $k_x^2 + 2k_y^2 = a^2$ and $q = 0$ along the lines $k_x = \pm\sqrt{2}k_y$. Four EP_3 s appear at $\{k_z = 0, k_x = \pm 1/\sqrt{2}, k_y = \pm a/2\}$. We identify two distinct situations.

(a) Adding a diagonal perturbation of the form $\text{diag}\{k_z, 0, -k_z\}$ (with $k_z \sim \epsilon$), the eigenvalues, upon moving out of the $k_z = 0$ plane, have a Puiseux series starting with the power law $k_z^{1/2}$. In particular, we find that the eigenvalues bifurcate as $\{0, \pm\sqrt{-1}\sqrt{k_z}(k_z - 2a)\}$ for the EP_3 at $\{k_z = 0, k_x = 1/\sqrt{2}, k_y = a/2\}$.

(b) Adding a PT-symmetry-preserving perturbation of the form

$$\epsilon \begin{bmatrix} 0 & ia_2 & \frac{(1-i)a_2}{\sqrt{2}} \\ ia_2 & 0 & 0 \\ \frac{(1+i)a_2}{\sqrt{2}} & 0 & 0 \end{bmatrix}, \quad (23)$$

with respect to the location of an EP_3 (which does not project out of the $k_z = 0$), we have an upper-3 Hessenberg matrix, $B_{3,3}$. It splits the threefold-degenerate zero eigenvalue into $\{\epsilon_3, \pm(-1)^{2/3}\epsilon_3\}$, where $\epsilon_3 = \sqrt[3]{aa_2^2\epsilon - 3a^2a_2}\epsilon^{1/3}$. Hence, it produces a leading-order splitting $\sim\epsilon^{1/3}$.

(ii) For $k_z = a$, we have $q = 0$ (identically) and $p = 0$ along the ellipse $k_x^2 + 2k_y^2 = 2a^2$. Here, we get EC_3 s along this elliptical curve. Adding a diagonal perturbation of the form $\epsilon \text{diag}\{1, 0, -1\}$, the eigenvalues, upon moving out of the $k_z = a$ plane, have a Puiseux series starting with the power law $\epsilon^{1/3}$. Needless to say, a generic PT-symmetric $B_{3,3}$ -type shows a leading-order splitting of the cube-root type. In particular, adding a PT-symmetry-preserving perturbation of the form

$$\epsilon \begin{bmatrix} 0 & 0 & (1-i)a_3 \\ 0 & 0 & (1+i)a_3 \\ (1+i)a_3 & -(1-i)a_3 & 0 \end{bmatrix}, \quad (24)$$

with respect to the location of the EC_3 , the threefold degenerate zero eigenvalue splits into $\{\epsilon_3, \pm(-1)^{1/3}\epsilon_3\}$, where $\epsilon_3 = 2^{5/6}\sqrt[3]{-a^2a_3(1 + \sin\alpha)}\cos\alpha\epsilon^{1/3}$.

V. DISCUSSIONS AND FUTURE OUTLOOK

In this paper, we have outlined an analysis to determine the order of singularities of exceptional degeneracies, given an s -dimensional complex square matrix. In particular, by determining its symmetry class [6], the kind of allowed symmetry-preserving perturbations (about the singular matrix at an EP) is constrained. On transforming an allowed perturbing matrix to the Jordan-normal basis of the defective matrix, its upper- ℓ Hessenberg structure (where ℓ is a positive integer $\leq s$) is revealed. It tells us the leading-order algebraic singularity (i.e., a branch point) that characterizes the corresponding EP, which also shows up as the leading-order fractional power, $1/\ell$, of the deviation-parameter (ϵ) away from the EP—the degenerate eigenvalues and eigenvectors splitting goes as $\epsilon^{1/\ell}$. After providing the generic prescription, we have moved onto apply the formalism to three-band models, considering three different symmetry classes (viz., P, C, and PT symmetries). We have proven that the most singular behavior that can appear in the P- and C-symmetric NH Hamiltonians is of the square-root variety. On the other hand, PT-symmetric NH Hamiltonians can generically host the strongest singular behavior possible for three-band systems, namely, $\epsilon^{1/3}$. Because of the spectator flat band in the three-band P- and C-symmetric systems (constrained to be present by the symmetries), the other two bands are negatives of each other, which immediately tells us that there is no possibility getting exceptional singularities of order 2. On the other hand, we do get EC_2 (singular curves) in the 3D PT-symmetric systems with three bands. Overall, we have demonstrated the emergence of singularities of all possible codimensions—EPs, ECs, and ESs—in the various systems studied here. Continuing our procedure for higher-band models will tell us the strongest algebraic singularities around their EPs. For example, four-band P- and C-symmetric 4×4

matrices allow EP_4 s which split as $\epsilon^{1/4}$ and $\epsilon^{1/2}$, respectively (see the Appendices). The curious behavior of the the latter originates from the Jordan-normal form comprising the Jordan blocks $\mathcal{J}_1(0)$ and $\mathcal{J}_3(0)$.

Through our mathematical analysis, we have been able to pinpoint the reason behind the singular behavior of a particular degree, emerging for a higher-order EP (or EC or ES) in a system protected by a specific symmetry. Furthermore, we have demonstrated that the perturbations can be fine-tuned such that the singular behavior is endowed with a directional dependence. This has been demonstrated in the (i) P- and C-symmetric systems whose energy-splittings can have either $\sim\epsilon$ (linear) or $\epsilon^{1/2}$ (square-root) nature; and (ii) PT-symmetric systems whose energy splittings can have either $\epsilon^{1/2}$ (square-root) or $\epsilon^{1/3}$ (cube-root) nature. This can be utilized in designing multifunctional EP-based sensors [35], which can now have direction-dependent sensitivity. We emphasize that the enhanced sensitivity is not fully contingent on exact degeneracy: it originates from the branch-point singularity of the eigenvalues (ϵ) in the vicinity of the EP. Since $\epsilon(\epsilon) - \epsilon(0) \sim \epsilon^{1/\ell}$, a sensor operated at a small but finite ϵ , where the system is nondegenerate, can still benefit from the enhanced spectral response associated with the nearby EP. We note, however, that enhanced spectral sensitivity alone does not necessarily imply enhanced measurement-precision, as the full sensing response is also influenced by eigenvector dynamics and noise properties [35,36].

While our derivation is presented using explicit matrix forms for transparency, the result admits a more general linear-algebraic interpretation. At an EP, the Hamiltonian is brought to its Jordan-normal form, whose generalized eigenvectors constitute a Jordan chain. A symmetry operator S does not act independently on the vectors in this chain. Rather, the symmetry relation, $SHS^{-1} = \eta H$ or $SH^T S^{-1} = \eta H$, depending on the symmetry class, requires every symmetry-preserving perturbation to belong to the corresponding intertwiner space of the Jordan representation. In the Jordan basis, this intertwining condition is expected to translate into linear constraints on the perturbation matrix elements, which we find to determine the admissible Hessenberg structure in the cases considered. For the P- and C-symmetry classes considered here, the symmetry operators square to the identity and induce a $\mathbb{Z}_2$ grading of the Hilbert space. The symmetry condition constrains both the Hamiltonian and every symmetry-preserving perturbation to respect this grading, yielding the characteristic block-off-diagonal structure in the natural basis. Upon transforming to the Jordan basis of the defective matrix, we find, consistent with our explicit construction, that these grading-constraints restrict the perturbation to the class of upper-2 Hessenberg matrices. By contrast, PT symmetry is implemented by the antiunitary constraint $H = (PK)H^*(PK)^{-1}$, which is an antilinear rather than a complex-linear condition on the Hamiltonian. Consequently, it does not induce an analogous $\mathbb{Z}_2$ grading of the complex vector space, and the corresponding intertwining constraints are correspondingly less restrictive. In the Jordan basis, they permit the larger class of upper-3 Hessenberg perturbations obtained in our explicit constructions. We thus expect the Hessenberg bandwidth to be governed by the structure of the symmetry-invariant intertwiner space associated

with the Jordan block, rather than by the particular matrix parametrization used in our calculations. We have verified this correspondence explicitly rather than in full generality. Finally, we note that our choice of symmetry-operator matrices is one convenient realization of the corresponding symmetry class. One can check explicitly that any other realization of the symmetry operators leaves both the Jordan-normal form and the symmetry constraints and, hence, the admissible perturbation space and the Hessenberg bandwidth unchanged.

Our terminology involving “topological phases” and “topologically robust phenomena” follows the established fact that nodal degeneracies are far more abundant and stable in NH systems than in their Hermitian counterparts, as reviewed in Ref. [3]. In a Hermitian system, a generic band-touching requires three real conditions to be satisfied simultaneously; therefore, such degeneracies are fine-tuned in two dimensions and become stable only in three dimensions. In NH systems, by contrast, a degeneracy is fixed by the vanishing of the complex discriminant of the characteristic polynomial, requiring only two real conditions. Therefore, EPs are already generic and stable in two dimensions, while still carrying a well-defined topological charge associated with spectral winding (vorticity) [37,38]. Recent work has further established EPs as topological singularities separating distinct NH spectral phases [18,39,40]. It is this reduced-codimension stability of EPs and, in the symmetry-constrained cases treated here, their additional protection under various symmetries (as established in our earlier work [4]) that motivate our terminology. We stress that this is a statement about the local stability and charge of the exceptional degeneracy itself, not about a global bulk invariant. A genuine bulk invariant, such as the biorthogonal polarization, is indeed computable in NH lattice models and has been shown to jump precisely at bulk phase transitions [18,39,40]. However, no such invariant is computed or compared across parameter regions in the present work, and we do not claim a conventional bulk topological phase transition. Last, but not least, emergence of EPs do indeed characterize phase transitions in systems hosting chiral Majorana zero modes [14,15].

In the future, it will be worthwhile to explore how to design a bidirectional sensor based on our theoretical observations. Another interesting direction is to investigate if and how the behavior of non-Hermitian skin effects (NHSEs) [16,19] change with the degree of the EP singularity and whether this can be traced back to the upper- ℓ value of the perturbing Hessenberg matrix. Attached to this is the question of the directional dependence of the NHSEs connected with the differing eigensystem-splitting, depending on the perturbing parameters, whether there are NH topological invariants to characterize the different scenarios [18,39–41], and whether the distinctions can be meaningfully quantified and identified in processes like topological pumping of charges [42,43].

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DATA AVAILABILITY

There are no publicly available research data or software supporting this manuscript. Requests for further information or data should be sent to the authors.

APPENDIX A: P-SYMMETRIC 4×4 MATRIX

We impose P symmetry on a 4×4 non-Hermitian matrix, which constrains its form to [4]

$$\mathcal{H}_4^P = \begin{bmatrix} 0 & 0 & a & b \\ 0 & 0 & c & d \\ e & f & 0 & 0 \\ g & h & 0 & 0 \end{bmatrix},$$

$$P = \text{diag}\{1, 1, -1, -1\},$$

$$P \mathcal{H}_4^P P^{-1} = -\mathcal{H}_4^P. \quad (\text{A1})$$

The 4×4 matrix P forms a four-dimensional realization of the P symmetry. For $a = bc/d$ and $g = -(ae + cf + dh)/b$, it hosts an EP_4 with the fourfold-degenerate eigenvalue $\lambda = 0$. At the defective point, its Jordan-normal form is

$$S^{-1} \mathcal{H}_{\text{EP}}^P S = \mathcal{J}_4(0), \quad \text{with}$$

$$S = \begin{bmatrix} 0 & -\frac{bd}{bce+cdf} & 0 & -\frac{bd^3h}{c^2(b,e+df)^2(cf+dh)} \\ 0 & -\frac{d^2}{bce+cdf} & 0 & -\frac{d^2[bce+d(cf+dh)]}{c^2(b,e+df)^2(cf+dh)} \\ -\frac{d}{c} & 0 & -\frac{d^2}{c^2(b,e+df)} & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

and $\mathcal{H}_{\text{EP}}^P = \mathcal{H}_4^P|_{a=\frac{bc}{d}, g=\frac{ae+cf+dh}{-b}}. \quad (\text{A2})$

Now let us add a P-symmetric perturbation, $\epsilon \tilde{B}_{4,4}^P$, to $\mathcal{H}_{\text{EP}}^P$, with

$$\tilde{B}_{4,4}^P = \begin{bmatrix} 0 & 0 & a_1 & a_2 \\ 0 & 0 & a_3 & a_4 \\ a_5 & a_6 & 0 & 0 \\ a_7 & a_8 & 0 & 0 \end{bmatrix}. \quad (\text{A3})$$

Explicit calculation shows that $S^{-1} \tilde{B}_{4,4}^P S$ is an upper-4 Hessenberg matrix, $B_{4,4}$, which we do not write here explicitly due to its lengthy expression. All the four eigenvalues behave as $\epsilon^{1/4}$, which we do not show here due to their complicated analytical forms.

APPENDIX B: C-SYMMETRIC 4×4 MATRIX

We impose C symmetry on a 4×4 non-Hermitian matrix, such that it gets constrained to the form,

$$\mathcal{H}_4^C = \begin{bmatrix} a & b & c & 0 \\ d & e & 0 & c \\ g & 0 & -e & b \\ 0 & g & d & -a \end{bmatrix},$$

$$C = \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix},$$

$$C(\mathcal{H}_4^C)^T C^{-1} = \epsilon_c \mathcal{H}_4^C$$

with $\epsilon_c = -1. \quad (\text{B1})$

The 4×4 matrix C forms a four-dimensional realization of the C symmetry with $\epsilon_c = -1$. For $c = -(a+e)^2/(4g)$ and $d = -(a^2 + 2cg + e^2)/(2b)$, it hosts an EP_4 with the fourfold-degenerate eigenvalue $\lambda = 0$. At the defective point, its Jordan-normal form is

$$S^{-1} \mathcal{H}_{\text{EP}}^C S = \begin{bmatrix} \mathcal{J}_1(0) & 0 \\ 0 & \mathcal{J}_3(0) \end{bmatrix}, \quad \text{with}$$

$$S = \begin{bmatrix} 0 & -\frac{b(a+e)^2}{2g} & 0 & 0 \\ \frac{(a+e)^2}{8bg} & \frac{(a-e)(a+e)^2}{4g} & -\frac{(a+e)^2}{4g} & 0 \\ \frac{1}{2} & -b(a+e) & b & 0 \\ \frac{e}{2b} & \frac{1}{2}(a-e)(a+e) & -a & 1 \end{bmatrix}$$

and $\mathcal{H}_{\text{EP}}^C = \mathcal{H}_4^C|_{c=-\frac{(a+e)^2}{4g}, d=-\frac{a^2+2cg+e^2}{2b}}. \quad (\text{B2})$

Intriguingly, the Jordan-normal form is split into two Jordan blocks, which needs special care. In the Jordan normal basis, we add a perturbation of the form

$$\mathcal{B} = \epsilon \begin{bmatrix} 0 & 0 \\ 0 & B_{3,2} \end{bmatrix}, \quad \text{setting } B_{3,2} = \begin{bmatrix} a_1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & -a_1 \end{bmatrix}, \quad (\text{B3})$$

where $B_{3,2}$ is a three-dimensional upper-2 Hessenberg matrix. This preserves the PT-symmetric form shown in Eq. (B1), because

$$S^{-1} \mathcal{B} S = \epsilon \begin{bmatrix} a_1 & 0 & 0 & 0 \\ \frac{a_1(e-a)+1}{2b} & 0 & 0 & 0 \\ \frac{2g[a_1(a+e)-1]}{(a+e)^2} & 0 & 0 & 0 \\ 0 & \frac{2g[a_1(a+e)-1]}{(a+e)^2} & \frac{a_1(e-a)+1}{2b} & -a_1 \end{bmatrix}. \quad (\text{B4})$$

Replacing $B_{3,2}$ by a $B_{3,3}$ matrix does not preserve the C-symmetric form and, hence, is not allowed. This simple example tells us that the eigensystem-splitting away from the EP_4 should be the same as the C-symmetric three-band models, namely, $\propto \epsilon^{1/2}$. Therefore, adding a C-symmetric perturbation with generic parameters, for example,

$$\epsilon \begin{bmatrix} a_1 & a_2 & a_3 & 0 \\ a_4 & a_5 & 0 & a_3 \\ a_6 & 0 & -a_5 & a_2 \\ 0 & a_6 & a_4 & -a_1 \end{bmatrix}, \quad (\text{B5})$$

produces eigenvalues behaving as $\epsilon^{1/2}$ (rather than $\epsilon^{1/4}$). We do not show here the closed-form analytical forms of the eigenvalues (appearing as $\pm$ pairs due to the C symmetry) because of their lengthy expressions.

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